

Wuao Liu

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RESEARCH INTERESTS	Computer vision and machine learning, with a focus on learning visual representations through multimodal supervision. Problems of interest include multimodal learning (audio-visual integration, vision and language), and video understanding. The application domains span AI for Science, AI for Robotics, etc.	
EDUCATION	University of Massachusetts Amherst <i>Ph.D. in Computer Science</i> Advised by: Dr. Grant Van Horn	Aug. 2024 – Apr. 2029 (expected) Amherst, MA
	University of Michigan <i>M.S. in Robotics</i> Advised by: Dr. Jason Corso	Aug. 2021 – Apr. 2023 Ann Arbor, MI
	Zhejiang University (ZJU) <i>B.Eng in Automation</i>	Sep. 2017 – Jun. 2021 Hangzhou, China
PUBLICATIONS	- Mustafa Chasmai, Wuao Liu, Subhransu Maji, and Grant Van Horn. Audio Geolocation: An Investigation with Natural Sounds. <i>In Submission</i>. - Filippos Bellos, Yayuan Li, Wuao Liu, and Jason Corso. Can Large Language Models Reason About Goal-Oriented Tasks? In <i>Proceedings of ACL Workshop on the Scaling Behavior of Large Language Models</i>, pages 24-34, 2024. [paper]	
WORK EXPERIENCE	The Corso Group <i>Research Engineer</i> Mentor: Dr. Jason Corso Project: Perceptually-enabled Task Guidance (PTG). We adopt foundation models (OpenAI Whisper, GPT models) for a VR system to assist humans in cooking. My responsibilities include training action recognition models and developing an Automatic Speech Recognition (ASR) module.	Jun. 2023 – Apr. 2024 Ann Arbor, MI
	Tencent AI Lab <i>Machine Learning Engineer Intern</i> Mentor: Dr. Peilin Zhao, Dr. Mingyang Sun Project: We developed a multi-agent reinforcement learning algorithm for optimizing electric vehicle charging and navigation systems using real-world data.	Jun. 2021 – Aug. 2021 Shenzhen, China
RESEARCH EXPERIENCE	Computer Vision Lab, UMass Advisor: Dr. Grant Van Horn Project: Contrastive Learning of Audio and Geographical Data - Proposed the first work to geolocate natural audio at the global scale. - Investigated the potential of species sounds as geolocation signals with strong oracles and proof of concepts. - Enabled multimodal (audio, image, text) geolocation with cross-modal contrastive learning.	Aug. 2024 – Dec. 2024 Amherst, MA

UM Ford Center for Autonomous Vehicles

May. 2022 – Aug. 2022

Advisor: Dr. Katie Skinner

Ann Arbor, MI

Project: **Neural Radiance Field (NeRF) for Autonomous Driving**

- Developed an open-source tool to collect RGB-D images in the CARLA simulator and visualize the corresponding camera poses.
- Implemented a scene-specific NeRF and achieved a PSNR of 28.29 dB for novel view synthesis in urban scenarios.
- Gained hands-on experience with multi-camera calibration using the Kalibr toolbox and worked with various types of cameras, such as event, RGB-D, and thermal cameras.

PRESENTATIONS

Audio Geolocation: An Investigation with Natural Sounds, New England Computer Vision (NECV) Workshop, Yale University. (November 2024)

**TECHNICAL
SKILLS**

- Programming Languages: Python, C/C++, MATLAB, L^AT_EX.
- Deep Learning: PyTorch, Tensorflow, Huggingface, Sklearn.
- Robotics Frameworks: ROS, GTSAM.